

Alison Miller  
DS 4002  
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## An Image Data Case Study: Exploring Movie Posters to Understand Trends on the Big Screen

### *Scenario:*

In July 2025, about 83% percent of Americans responded that they watch streaming services like Netflix, Paramount+, and others. As streaming continues to rise in popularity, it is reasonable to assume the financial backers of the TV and film industry are taking notes on what is popular on their platforms to inform their broader strategy. Across platforms, we can see that the most popular shows are those that feature an ensemble cast as opposed to just one major star. The ensemble cast is not a new idea in film media but in recent years it seems to have found a recent resurgence in TV series format particularly. With many of the most successful new original series on streaming platforms focusing on narrative through strong ensembles of actors instead of one dominant star, think *White Lotus* and *Stranger Things*, it leaves one wondering if this trend has been embraced at the box office and in streaming native movies? Through this case study, we aim to study the nature of the featured person count and popularity in movies through movie posters. It is a reasonable assumption that ensemble based movies feature an increased number of persons on their poster given a marketing perspective.

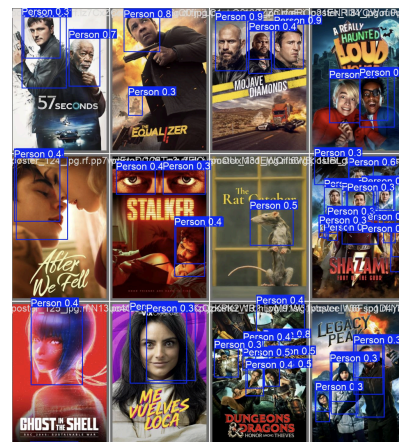
Now the question is, how do we prove or disprove this potential phenomena using data science methods around image data?

### *Deliverable:*

Our goal with this case study is to conduct person detection on these posters using both a pretrained and custom trained model from Ultralytics. Hand counting people in a large movie poster data set is laborious so we want to use our data science skills to help us. After working through using the two iterations of the model, we will apply some of our math and coding skills to perform an analysis between the person in the poster and the given popularity score for the movie in the data set. Your deliverables will be to utilize both models to count persons in a poster and this mathematical trend analysis. A project like this can be thought of as creating a trend report to help understand if streaming native series are in the proverbial drivers seat for new releases on the big screen today!



*Example Movie Poster.*



*Example results from Person Detection.*